And on the other hand, an application of the continuous mapping theorem yields

$$\begin{aligned}
&G \circ F_2\big([\sqrt{m}(\widehat{\sigma}^2_{\mathcal{X}}(s,t) - st\sigma^2_{\mathcal{X}})]_{s,t}, [\sqrt{n}(\widehat{\sigma}^2_{\mathcal{Y}}(s,t) - st\sigma^2_{\mathcal{Y}})]_{s,t}\big) \\
&\quad \Rightarrow G\Big(F_2\big([tW_{\Gamma_X}(s) + sW_{\Gamma_X}(t)]_{s,t}, [tW_{\Gamma_Y}(s) + sW_{\Gamma_Y}(t)]_{s,t}\big)\Big) \\
&\quad \overset{\mathcal{L}}{=} \left\{\xi^2 \int_{[0,1]^2} \Big[st\big(tB_1(s) + sB_1(t) - 2stB_1(1)\big)\Big]^2 \nu(\mathrm{d}(s,t))\right\}^{1/2}.
\end{aligned}$$

Step 4. Finally, the map

$$D_2 \times D_2 \to \mathbb{R}^2, \quad (u,v) \mapsto \begin{pmatrix} F_1(u,v) \\ G \circ F_2(u,v) \end{pmatrix}$$

is continuous w.r.t. the product metrics and the convergence in probability shown in (3.27) and (3.29) carries over to the two-dimensional setting as well. This completes the proof. ☐

3.3. Functional CLT for U-statistics

Proof of Theorem 2.4. Let $g\colon (D_1, \mathfrak{D}_1) \to (D_2, \mathfrak{D}_2)$ be defined by $g(x)(s,t) := tx(s) + sx(t)$, $s,t \in [0,1]$. Consider the process

$$X_n\colon [0,1] \to \mathbb{R}, \quad t \mapsto \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} Z_i$$

Then $Y_n = g(X_n)$ and $X_n \Rightarrow \mathcal{W}_\Gamma$ in $(D_1, \mathfrak{D}_1)$ by the functional central limit theorem (Theorem A.1) given in Aue et al. [1]. Thus it remains to verify that $g\colon D_1 \to D_2$ is continuous with respect to the Skorokhod metrics ($q = 1,2$)

$$d_q(x,y) := \inf_{\lambda \in \Lambda_q} \left\{\max\left\{\sup_t |x(t) - y(\lambda(t))|, \sup_t \|\lambda(t) - t\|\right\}\right\}, \quad x,y \in D_q, \tag{3.30}$$

where the infimum is taken over the class Λ_q of all "time transformations" $\lambda(t) = (\lambda_1(t_1), \dots, \lambda_q(t_q))$ such that $\lambda_r\colon [0,1] \to [0,1]$ is continuous and strictly increasing with $\lambda_r(0) = 0$ and $\lambda_r(1) = 1$.
Now, assume that $x,y \in D_1$ with $d_1(x,y) < \delta$. Then there is a $\lambda^* \in \Lambda_1$ such that

$$\sup_t |x(t) - y(\lambda^*(t))| < \delta \quad \text{and} \quad \sup_t |\lambda^*(t) - t| < \delta.$$

Put $\widetilde{\lambda} = (\lambda^*, \lambda^*) \in \Lambda_2$, then

$$|g(x)(s,t) - g(y)(\widetilde{\lambda}(s,t))|$$

$$\begin{aligned}
&=|tx(s) \mp \lambda^*(t)x(s) - \lambda^*(t)y(\lambda^*(s)) + sx(t) \mp \lambda^*(s)x(t) - \lambda^*(s)y(\lambda^*(t))| \\
&\leq |t - \lambda^*(t)||x(s)| + |\lambda^*(t)||x(s) - y(\lambda^*(s))| + |s - \lambda^*(s)||x(t)| \\
&\quad + |\lambda^*(s)||x(t) - y(\lambda^*(t))| \\
&\leq 2(1 + \sup|x|)\delta
\end{aligned}$$

as well as

$$\|\widetilde{\lambda}(s,t) - (s,t)\| = |\lambda^*(s) - s| + |\lambda^*(t) - t| < 2\delta.$$

Now, let $\varepsilon > 0$ and $x \in D_1$ be arbitrary but fixed. Set $\delta \coloneqq \varepsilon/[2(1 + \sup|x|)]$. Then we may conclude for all $y \in D_1$ with $d_1(x,y) < \delta$, that

$$d_2(g(x), g(y)) < 2(1 + \sup|x|)\delta = \varepsilon.$$

This shows the continuity of g and the assertion immediately follows from the continuous mapping theorem, i.e., $Y_n = g(X_n)$ converges weakly to $g(X) = Y$. $\square$

Proof of Theorem 2.5. The proof is divided into several steps.

(1) Reduction to equidistant grid. Firstly, we argue that it is sufficient to consider blocks $A = (s,u] \times (t,v]$ with corner points $(s,t), (u,v)$ in the equidistant grid Π_n of $[0,1]^2$, where the grid is given by

$$\Pi_n = \Big\{ \Big(\frac{i}{n}, \frac{j}{n}\Big) : 0 \leq i, j \leq n \Big\}.$$

Let $(s,t) \in [0,1]^2$ and define $\underline{s} = \lfloor ns \rfloor / n$ and $\underline{t} = \lfloor nt \rfloor / n$. Then $(\underline{s}, \underline{t})$ is the point on the grid Π_n which satisfies $U_n(h_2)(s,t) = U_n(h_2)(\underline{s}, \underline{t})$. In particular, setting $\underline{A} = (\underline{s}, \underline{u}] \times (\underline{t}, \underline{v}]$ for $A = (s,u] \times (t,v]$ yields immediately $U_n(h_2)(A) = U_n(h_2)(\underline{A})$. So, in the following, we can assume that $A = (s,u] \times (t,v]$ for $(s,t), (u,v) \in \Pi_n$. In particular, we will exploit that in this case both $t - s, v - u \geq n^{-1}$.

(2) Position of the block. Secondly, we argue that it is sufficient to study two special types of blocks A on the grid Π_n only:

(a) $A = (s,u] \times (t,v]$ with $s < u \leq t < v$, is termed a block "above the diagonal";
(b) $A = (s,u] \times (s,u]$ with $s < u$, is termed a block "on the diagonal".

Indeed, every block $A \subset [0,1]^2$ which is not "on the diagonal" can be split into four blocks A_1, A_2, A_3, A_4 such that exactly one block, A_1 say, is of type (b) ("on the diagonal") and the remaining three blocks, A_2, A_3, A_4, are either of type (a) ("above the diagonal") or are of type (a) when being "transposed". By the latter we mean that the transpose of a block $B = (s,u] \times (t,v]$ is $B^T = (t,v] \times (s,u]$, i.e., the block B is reflected along the diagonal. Due to the symmetry of the functional h_2 we have that the increment over a block coincides with the increment over the corresponding transposed block, i.e., $U_n(h_2)(B) = U_n(h_2)(B^T)$, and hence we can assume w.l.o.g. that A_2, A_3, A_4 are all of type (a).

Having made this partitioning of a block A which is "not on the diagonal", observe that by definition of the 2-dimensional increment, common corner points of neighboring blocks cancel each other and therefore $U_n(h_2)(A) = \sum_{i=1}^4 U_n(h_2)(A_i)$. Applying the Cauchy-Schwarz inequality yields

$$\mathbb{E}[|\sqrt{n}U_n(h_2)(A)|^2] \leq 4\sum_{i=1}^{4} \mathbb{E}[|\sqrt{n}U_n(h_2)(A_i)|^2]. \tag{3.31}$$

Thus, it remains to proof (2.14) for blocks A of type (a) or type (b), because then - as an immediate consequence of (3.31) - we conclude

$$\mathbb{E}[|\sqrt{n}U_n(h_2)(A)|^2] \leq 4C \sum_{i=1}^{4} |A_i|^{3/2} \leq 4C|A|^{3/2}.$$

(3) Hölder continuity. We use the Hölder continuity of h_2 (see (2.10)) in combination with the L^p-m-property of the process to obtain continuity properties which are necessary for the moment bound. To this end, let $Y_1, \ldots, Y_4$ and $Z_1, \ldots, Z_4$ be M-valued random elements with marginal distribution equal to $\mathbb{P}_X$ and an arbitrary joint distribution. Then

$$\begin{aligned}
&|\mathbb{E}[h_2(Y_1,Y_2)h_2(Y_3,Y_4)] - \mathbb{E}[h_2(Z_1,Z_2)h_2(Z_3,Z_4)]| \\
=&|\mathbb{E}[h_2(Y_1,Y_2)(h_2(Y_3,Y_4) - h_2(Z_3,Z_4))] \\
&+ \mathbb{E}[h_2(Z_3,Z_4)(h_2(Y_1,Y_2) - h_2(Z_1,Z_2))]| \\
\leq&\mathbb{E}[|h_2(Y_1,Y_2)|^q]^{1/q}\ \mathbb{E}[|h_2(Y_3,Y_4) - h_2(Z_3,Z_4)|^p]^{1/p} \\
&\quad + \mathbb{E}[|h_2(Z_3,Z_4)|^q]^{1/q}\ \mathbb{E}[|h_2(Y_1,Y_2) - h_2(Z_1,Z_2)|^p]^{1/p} \\
&\leq 3L \sup_{U,V\sim\mathbb{P}_X} \mathbb{E}[|h_2(U,V)|^q]^{1/q}\Big\{\mathbb{E}[(d(Y_3,Z_3)+d(Y_4,Z_4))^{\rho p}]^{1/p} \\
&\quad + \mathbb{E}[(d(Y_1,Z_1)+d(Y_2,Z_2))^{\rho p}]^{1/p}\Big\} \\
&\leq 3L \sup_{U,V\sim\mathbb{P}_X} \mathbb{E}[|h_2(U,V)|^q]^{1/q}\Big\{\mathbb{E}[d(Y_3,Z_3)^{\rho p}]^{1/p} + \mathbb{E}[d(Y_4,Z_4)^{\rho p}]^{1/p} \\
&\qquad\qquad + \mathbb{E}[d(Y_1,Z_1)^{\rho p}]^{1/p} + \mathbb{E}[d(Y_2,Z_2)^{\rho p}]^{1/p}\Big\} \\
&= 3L \sup_{U,V\sim\mathbb{P}_X} \mathbb{E}[|h_2(U,V)|^q]^{1/q} \sum_{i=1}^{4} \mathbb{E}[d(Y_i,Z_i)^{\rho p}]^{1/p},
\end{aligned} \tag{3.32}$$

where $p, q \geq 1$ are Hölder conjugate and where the second inequality is a consequence of (2.10); the last inequality is a consequence of the Minkowski inequality and the fact that $(a+b)^\rho \leq a^\rho + b^\rho$ for $a, b \in \mathbb{R}_+$.

Furthermore

$$\mathbb{E}[|h_2(U,V)|^q]^{1/q} \leq \mathbb{E}[|h(U,V)|^q]^{1/q} + \theta + \mathbb{E}[|h_1(U)|^q]^{1/q} + \mathbb{E}[|h_1(V)|^q]^{1/q}$$